

The original Fourier Hessian, all 289 rows

$$K(\omega) = \mathcal{H}_{\text{original}}(\omega), \quad G(\omega), \quad E = -\text{color}_2 - Y/2$$

Original physical frequency

$$R_\omega = \text{Res}_{\omega = \omega^*} G(\omega)$$

$$K'_\omega = \partial_\omega K(\omega^*)$$

$$P = R_\omega K'_\omega$$

Clock re-expression, positive scale

$$t' = ct, \quad \omega' = \omega/c, \quad c > 0$$

$$R_c = R_\omega/c, \quad K'_c = cK'_\omega$$

$$R_c K'_c = P$$

The flux identity returns with the same clock

$$\mathcal{E}^T R_c K'_c R_c \mathcal{E} = \mathcal{E}^T R_c \mathcal{E}$$

Both sides scale by $1/c$; the normalized projector P is retained